Numerical Simulation Methods And Their Applications

Monte Carlo Methods and Applications in Particle-based Systems

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Plain language summary

This project concerns working with a computer simulation of a particle-based system, particularly using numerical techniques known as Monte Carlo simulations.

It starts with an introduction to the history of these methods, touching upon the infamous Manhattan Project, then focusing on pseudo-random number generators, sampling methodologies, and their applications in statistical mechanics. The project also explores the Metropolis method and its application to a Lennard-Jones liquid. Finally, it concludes with findings of a simulation created from individually developed C++ code.

Declaration

I certify that this is my own work and that use of material from other sources has been properly and fully acknowledged in the text. I have read the University's definition of plagiarism and the department's advice on good academic practice. I understand that the consequence of committing plagiarism, if proven and in the absence of mitigating circumstances may include failure in the Year or Part or removal from the membership of the University. I also certify that neither this piece of work, nor any part of it, has been submitted in connection with another assessment.

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1 Development Of Computer Simulation Methods

1.1 The Manhattan Project

1.1.1 Background

Before understanding Monte Carlo Methods, and the higher level category of "Computer Simulations", we are bound to mention **The Manhattan Project**. The top-secret research and development program was the first to produce Nuclear Weapons, and was a major playing card in the context of the greater geopolitical climate: World War II.

Funded by nations such as the UK, Canada and, majorly, the USA, the Manhattan Project is often cited as a major point in human history. Disregarding moral arguments concerning atomic weaponry, a large factor in the awe of the project was the pure advancement in scientific knowledgement - building upon previous discoveries of fission.

1.1.2 Stanisław Ulam, Von Neumann and Metropolis

1.1.3 Subsequent Developments

2 Monte Carlo Methods

2.1 Pseudo-Random Number Generators

Random number generation is an important part of Monte Carlo methods. There are multiple different methods to obtaining random numbers, but the algorithms we are focusing on within this report are called **Pseudo-Random Number Generators** (PRNGs). While they do not generate "actual" random numbers, what they generate is sufficient enough in variability that it can be used in scenarios where random numbers are needed.

The way that they achieve this is through the use of mathematical formulae, varying in levels of complexity. The algorithms can get increasingly elaborate, yet the simplest can consist of only a few lines of code utilising basic mathematical arithmetic. Sequences are generated which attempt to approximate random numbers, based on an initial starting *seed*.

While humans can not predict the next number in the sequences that are generated, this is not perfectly computationally secure. This is because if the initial *seed state* is known, an identical sequence can be reproduced. Therefore, the sequences are deterministic and not as unpredictable as systems with true chaos built into them - such as radioactive atom decay, or, famously, *Cloudflare*'s wall of lava lamps (Liebow-Feeser 2017).

2.1.1 Mersenne Twister

2.1.2 Well Equidistributed Long-period Linear (WELL)

2.2 Sampling Methodology

2.2.1 Simple

2.2.2 Importance

$I = \int_a^b \ f(x) \ dx$

2.3 Statistical Mechanics

2.4 The Metropolis Method

3 Question 3? Lennard-Jones ?

4 Practical Application in a Numerical Simulation

4.1 Code and development

4.2 Application to a Lennard-Jones liquid

4.2.1 Numerical Analysis

5 Conclusions, Findings, Prospectives

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